

Lunar Regolith Database User Manual

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Contents

1 Overview	2
2 Quick Start	2
2.1 Web application	2
2.2 Local Database Script	2
3 System Requirements	2
4 Database Content	3
4.1 Database Pages	3
4.2 Parameters	3
5 Database Structure	4
5.1 Database Table	4
5.2 Sidebar	4
5.2.1 Pages	4
5.2.2 Filters	5
5.2.3 Table Columns Display	6
5.3 Plotting Data	6
5.4 Lunar Map and Missions Representation	6
6 Debugging	7
6.1 Installation errors	7
6.2 Last commit local error	8
7 Contact	8

1 Overview

The Lunar Regolith Database is an interactive tool that allows users to explore, visualize, and compare lunar regolith properties measured during space missions, as well as properties of lunar regolith simulants. Users can filter and sort the data, and generate different types of plots for analysis.

A publication detailing the database contents and some possible use cases is available at <https://arxiv.org/abs/2602.03829> [1]. The dataset used in the most recent version of the database is available at <https://doi.org/10.57745/NTSZ8G> [2].

This user manual describes the requirements, features, and functions of the *Lunar Regolith Database*.

2 Quick Start

2.1 Web application

Access the database through the streamlit app with the link: <https://lunar-regolith-database.streamlit.app/> or through the hugging face space with the link: <https://huggingface.co/spaces/leoniegasteiner/Lunar-Regolith-Database>. If the app is asleep, just click on the button "get this app back up" and wait until the display appears. This step requires a connection to the internet.

2.2 Local Database Script

Follow these steps to install the Lunar Regolith Database on your device and be able to run it without internet connection:

1. **Install Python** (version 3.9–3.12 recommended).

2. **Download the project:**

```
git clone https://github.com/leoniegasteiner/Lunar-Regolith-Database
cd <leoniegasteiner/Lunar-Regolith-Database>
```

or download the ZIP file, unzip it, and open the folder in your Python editor.

3. **Install required packages:**

```
pip install streamlit pandas plotly numpy openpyxl matplotlib
```

Note: The streamlit package is not compatible with Python versions above 3.12.

4. **Run the database:** To run the database, click on the app icon or type the following command in your terminal:

```
streamlit run Lunar_Database.py
```

5. **Open the interface:** A local link will appear in the terminal. Click it, or copy and paste it into Chrome, Firefox, Safari, or Edge. The database window should open automatically in a new tab.

That's it - the database is ready to use. See the following sections of this manual for details on filtering, plotting data, and using the lunar map.

3 System Requirements

The database is written as a Python script that runs with the *Streamlit* visualization package. When executed, Streamlit opens an interactive browser window for display. For this reason, users must have Python installed, along with the required packages.

The system requirements to run the database are:

- **Python:** Version 3.9 to 3.12 (recommended)
- **Operating system:** Windows, macOS, or Linux
- **RAM:** Minimum 4 GB
- **Internet browser:** Chrome, Firefox, Safari, or Edge

4 Database Content

The database contains data collected from different mission reports and articles from lunar missions and on lunar regolith samples and simulants testing. Data extraction was performed manually to ensure accuracy, given the often qualitative and sometimes unstructured nature of original sources. The database will continue to be updated over time, reflecting new findings and contributions.

4.1 Database Pages

The data collected is displayed in different pages accessible through the sidebar:

- the *Lunar Regolith Database* page contains the data collected during the different lunar missions, including the data from *in-situ* and on Earth testing, and remote sensing data;
- the *Lunar Regolith Simulants Database* page contains data on the different lunar regolith simulants;
- the *Lunar Samples Database* page contains a table with information on over 2000 lunar samples brought back to Earth. It also contains a table with the particle size distribution and median grain size of over 300 samples. A plotting section allows to plot the cumulative PSD of different sample or to generate a scatter plot of the median particle size as a function of the depth of the sample collection.
- the *Mission Details* page contains more information on the context of each mission and the tests performed *in-situ*, on earth, and/or remotely;
- the *Combined Data* page contains a large table with all the data we collected on lunar regolith and its simulants.

4.2 Parameters

To accurately reflect geomechanical testing principles, the database categorizes parameters into metadata, direct field measurements, laboratory testing conditions, and derived geomechanical properties. The current version of the database includes (but is not limited to):

Metadata

- **Mission, Location, Terrain, and Year:** parameters specific to the mission considered, with the location of the spacecraft landing, the geologic terrain of the region explored, and the year of the mission;
- **Test:** the measurement method used;
- **Test location:** indicates whether the measurement was performed *in-situ* on the lunar surface, on Earth using returned samples, or remotely for remote sensing data;
- **Testing environment:** specifies the atmosphere in which the test was performed; in vacuum for *in-situ* and remote measurements, and in vacuum, nitrogen, or air for returned samples;
- **Sample ID:** specifies the identification number of the sample on which the test was performed, when applicable;
- **Reference and Comments:** bibliographic citation of the source from which the data was extracted and additional information relevant to the specific entry line.

Direct Field Measurements

- **Depth (cm):** depth below the lunar surface at which the measurement was taken, when available;
- **Force applied (N):** for penetrometer tests, the force applied on the soil during the test, when available;
- **Static bearing pressure (kPa):** the estimated pressure exerted on the regolith by lander footpads, astronaut footprints, or rover wheels based on their respective weights in lunar gravity;
- **Contact area (cm):** area over which the applied force or static bearing pressure is applied.

Laboratory Testing Conditions

- **Normal stress range (kPa):** range of normal stresses applied during shear testing of returned samples on earth from which the Mohr-Coulomb parameters were derived;

Derived and Inferred Mechanical Properties

- **Angle of internal friction (degree):** first parameter in the Mohr-Coulomb model, corresponding to the slope of the Coulomb failure line;
- **Cohesion (kPa):** second parameter in the Mohr-Coulomb model, corresponding to the intercept of the Coulomb failure line with the shear strength axis;
- **Bulk density (g/cm³):** defined as the mass of unconsolidated regolith per unit volume;
- **Bearing capacity (kPa):** defined as the maximum load per unit area that the regolith can support without failure;
- **Cone penetration resistance gradient (kN/m²/m):** the rate at which the soil's resistance to penetration increases with depth, calculated from the applied force and penetrometer geometry;
- **Porosity (%) and Void ratio:** interconvertible expressions of the volume of voids relative to the total or solid volume of the soil;
- **Density of grains (g/cm³):** the mass of the solid soil particles per unit volume of solids;
- **Compressibility coefficient:** the soil's volume change behavior under applied normal stress.

The physical parameters included in the database were extracted from multiple sources, beginning with the earliest available reports and incorporating subsequent interpretations when possible. When several values were reported for the same parameter, mission, and testing method, all were retained to provide a comprehensive overview of the data. The database will continue to be updated over time, reflecting new findings and contributions.

5 Database Structure

5.1 Database Table

The database table (Figure 1) appears automatically below the title of the database page selected. Each column can be sorted by clicking its header, and data can be explored by scrolling vertically or horizontally. The table can be expanded to full screen by clicking the full screen icon that appears in the top right corner when hovering the mouse over the table.

The table can also be downloaded in .csv format by clicking the download icon (top right corner). The exported file will reflect any filters or sorting currently applied.

The parameters displayed in the table can be modified using the *Select Table Columns* section in the sidebar.

5.2 Sidebar

The sidebar in the Database allows to select the database page, filter the data, and select the parameters displayed in the table.

5.2.1 Pages

The database contains multiple pages, each page can be accessed through the sidebar in the area shown in Figure 2.

Database Table

Search

Download as csv

Fullscreen

	Mission	Location	Terrain	Year	Type of mission	Test	
0	Surveyor I	2.4745 43.339W	Mare	1966	Lander	Spacecra	In-Situ
1	Surveyor I	2.4745 43.339W	Mare	1966	Lander	Spacecra	In-Situ
2	Surveyor I	2.4745 43.339W	Mare	1966	Lander	Spacecra	In-Situ
3	Surveyor I	2.4745 43.339W	Mare	1966	Lander	Spacecra	In-Situ
4	Surveyor III	3.05 23.41W	Mare	1967	Lander	Spacecra	In-Situ
5	Surveyor III	3.05 23.41W	Mare	1967	Lander	Spacecra	In-Situ
6	Surveyor III	3.05 23.41W	Mare	1967	Lander	Spacecraft touchdown analysis	In-Situ
7	Surveyor V	1.42N 23.20E	Mare	1967	Lander	Spacecraft touchdown analysis	In-Situ
8	Surveyor VI	0.473N 1.427W	Mare	1967	Lander	Spacecraft touchdown analysis	In-Situ
9	Surveyor VI	0.473N 1.427W	Mare	1967	Lander	Vernier engine	In-Situ

Column sorting menu

Figure 1: Table interface and functions details

Select Database:

☒ Lunar Regolith Database

☐ Lunar Regolith Simulants Database

☐ Lunar Samples Database

☐ Detailed Mission Pages

☐ Combined Data

Figure 2: Sidebar page selection

Mission Group

Select Mission Group

Clear element

Clear all selected filter elements

Apollo x Chang'e x

Luna

Surveyor

Chandrayaan

Lunar Reconnaissance Orbiter

Other

Non selected filter elements

Cohesion (kPa)

Select Cohesion Range

2.69 11.76

Figure 3: Categorical & Numerical filter

5.2.2 Filters

The content of both the table and the plots can be restricted using the filter section in the sidebar (Figure 3). There are two kinds of filters, each in a separate tab for display purposes:

- **Categorical filters:** Allow to discriminate categorical parameters (e.g. type of terrain, mission group, etc.).
- **Numerical range filters:** Allow to specify a minimum and maximum value for numerical parameters (e.g. year, cohesion, etc.).

Multiple filters can be applied simultaneously. A filter can be removed by clicking the small cross next to its name, and all filters can be cleared at once by clicking the *Clear all filters* button at the bottom of the sidebar.

The selected filters determine the lines displayed in the table and the data points appearing in the plots.

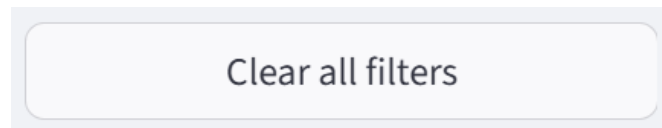


Figure 4: Clear all filters button

5.2.3 Table Columns Display

At the bottom of the sidebar, a filter allows to select which columns and parameters are displayed in the table. This allows to tailor the table to the user's need. A set of default parameters is automatically displayed when the database is opened, additional parameters and details can be added using this display option. The selection and removal of columns functions in the same way as for the categorical filters.

The *Select All Parameters* button enables the user to display all possible available parameters without the need of individually selecting each of them, and the *Clear selection* allows to go back to default table display.

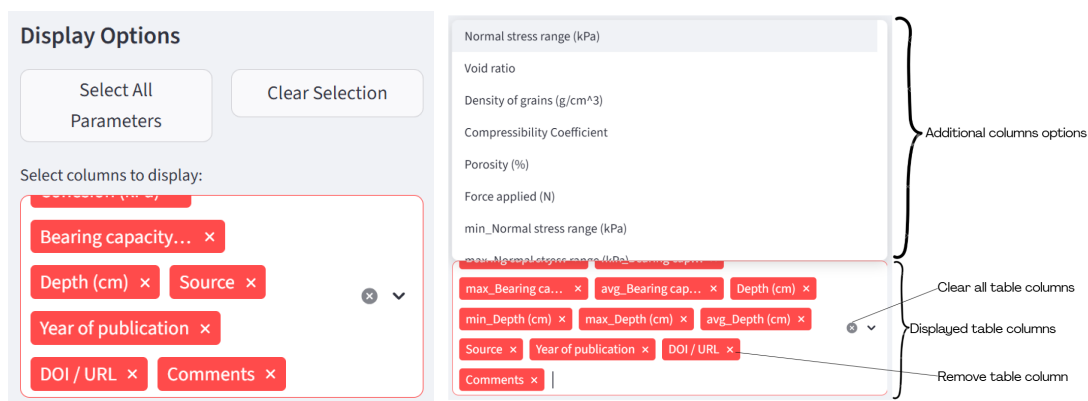


Figure 5: Sidebar columns display selection

5.3 Plotting Data

The *Plot Numerical Data* section (Figure 6) allows users to visualize the filtered data in different plots. The X- and Y-axes can be selected from the drop-down menus above the plot. The format of the plot can be selected among the options of *Range*, *Average*, *Minimum*, and *Maximum*. The Range mode will plot the data as ranges, having between 0 and 2 values per parameter. The Average, Minimum, and Maximum options are displayed as scatter plots.

Data points are grouped by different legends that can be selected (e.g. mission group, terrain, etc.) represented with different marker shapes and colors. The legend on the right allows interactive filtering: clicking a group name hides its points, while double-clicking a group isolates it by hiding all others.

In the *Lunar Regolith Database* page, if the plotted parameters are available in the lunar regolith simulants dataset, a checkbox appears below the plot settings. This enables comparison with lunar regolith simulants: when selected, the corresponding simulant data points are added to the plot.

5.4 Lunar Map and Missions Representation

At the bottom of the page, the database includes a lunar surface map (Figure 7), where the landing sites of missions are represented, distinguished depending on the legend group selected in the plotting section. Each legend group is assigned a specific color and marker shape. As with the plots, legend groups can be toggled on or off in the legend to focus on selected data.

Plot Numerical Data

X-axis (categorical)

Mission

X-axis parameter selection (Categorical or Numerical)

Y-axis (numeric)

Bulk density (g/cm³)

Y-axis parameter selection (Numerical)

Select value type to plot

☒

Range

☐

Average

☐

Minimum

☐

Maximum

Plotting mode selection (Value ranges, average values, minimum values, or maximum values)

Select Legend/Color Grouping

Mission Group

Legend group selection

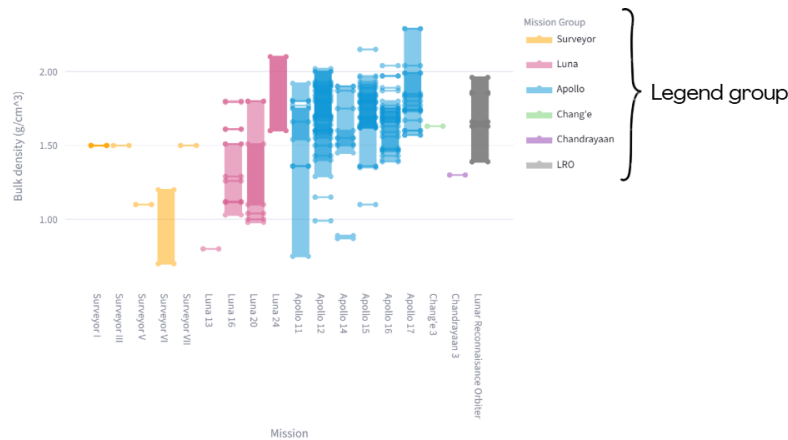
Bulk density (g/cm³) Range vs Mission Range

Figure 6: Plotting visualization and explanations

Mission Location Representation on the Moon (Filtered by Mission Group)

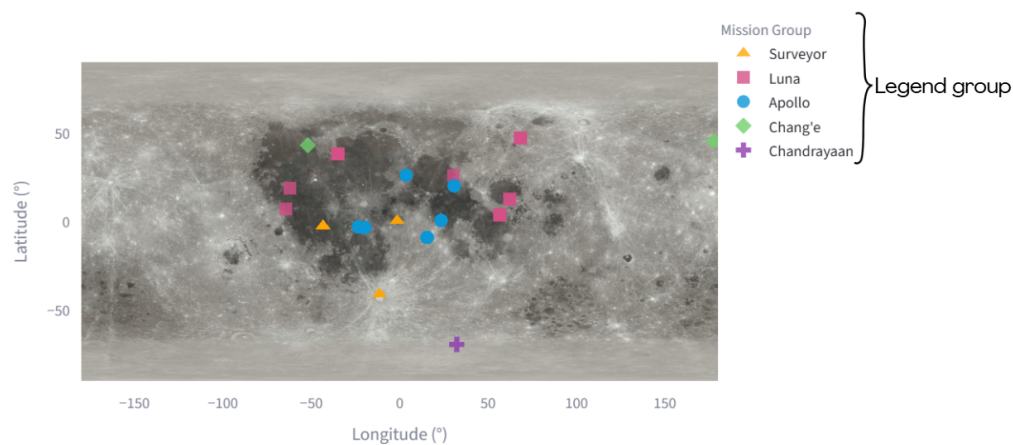


Figure 7: Lunar map visualization and explanations

6 Debugging

6.1 Installation errors

This section lists common issues and their solutions. In case of persistent malfunction of the database, please contact us at leonie.gasteiner@student.isae-superaero.fr

- **Streamlit does not start:** Make sure you installed all dependencies and are using Python 3.9–3.12.
- **Page does not open in browser:** Copy the local link displayed in the terminal and paste it into a supported browser (Chrome, Firefox, Safari, Edge).
- **Plots do not appear:** Check that your filter settings do not exclude all available data.
- **Module not found errors:** Reinstall dependencies with `pip install -r requirements.txt` or install the missing package individually.

6.2 Last commit local error

Since the folder downloaded to run the database locally is the same as the one running the web application, some functionalities are not available when running the script locally. Notably, the footer might display an error, this should not affect any other part of the database but can be removed for aesthetic purposes:

1. Open the script titled *Lunar_Database.py* and scroll to the very end;
2. Comment out the content from line 2600 to 2625 using `'''` before and after the paragraph, or `#` at the beginning of each line

7 Contact

Don't hesitate to contact us in case of persistent bugs, or to submit new data or display/format requests.

Email: leonie.gasteiner@student.isae-supaero.fr

References

- [1] L. Gasteiner, N. Murdoch, and O. D'Angelo, "An Open Database of Lunar Regolith and Simulants Properties," 2026. invited contribution for the Journal for Numerical and Analytical Methods in Geomechanics [preprint: ArXiv: 2602.03829].
- [2] L. GASTEINER, N. MURDOCH, and O. D'ANGELO, "Dataset for the Lunar Regolith Database," May 2026. <https://entrepot.recherche.data.gouv.fr/dataset.xhtml?persistentId=doi:10.57745/NTSZ8G>.